OPERATING SYSTEM LAB

COMMAND REFERENCE

Introduction:

The Linux operating system is a widely used open-source platform known for its flexibility, security, and powerful command-line interface. It is commonly used in system administration, software development, and file management. This lab experiment was conducted to gain practical knowledge of various Linux commands in the Ubuntu environment. The tasks included file and directory operations, permission management, system monitoring, and compression techniques. Through this hands-on practice, the functionality and real-life use of Linux commands were better understood.

SECTION 1: File System Navigation & Directory Commands

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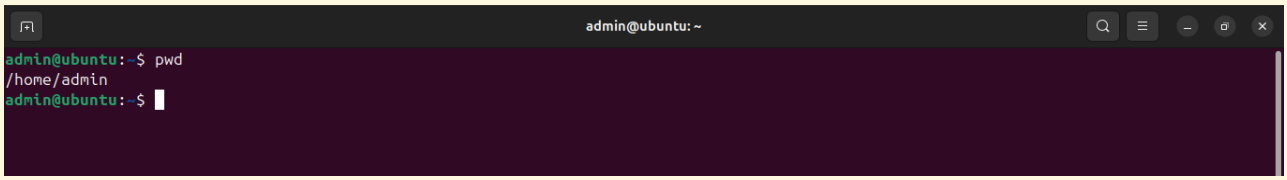
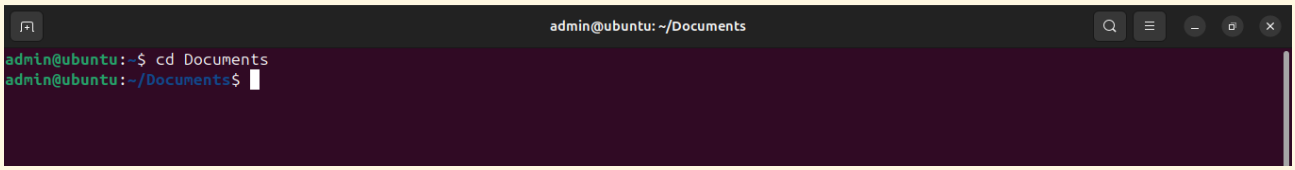
Command 1: pwd

Syntax	`pwd`
Description	Prints the present (current) working directory. It shows the full absolute path from the root (/) to the current directory. Very useful when navigating deep directory structures.



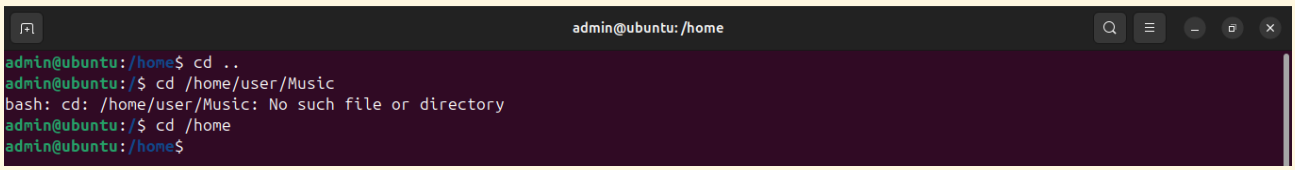
Command 2: cd

Syntax	`cd [directory]`
Description	Changes the current working directory. With no argument, it returns to the home directory. Used constantly in terminal navigation.



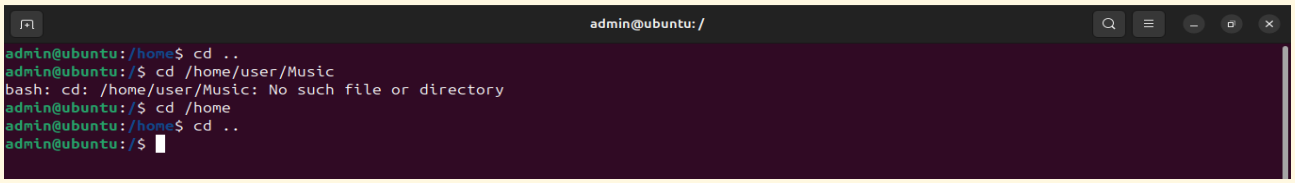
Command 3: `cd /home/user/Music`

Syntax	`cd /home/user/Music`
Description	Changes directory using an absolute path. This navigates directly to /home/user/Music regardless of current location. Absolute paths always start from root (/).



Command 4: `cd ..`

Syntax	`cd ..`
Description	Moves one level up in the directory hierarchy (to the parent directory). Two dots (..) always refer to the parent of the current directory.



Command 5: `ls`

Syntax	`ls [options] [path]`
Description	Lists the contents (files and folders) of the current directory. Without options, displays names only. One of the most frequently used Linux commands.

```
admin@ubuntu:~$ cd Documents
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          randomtxt.txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  script.sh
admin@ubuntu:~/Documents$
```

Command 6: `ls -l`

Syntax	`ls -l [path]`
Description	Lists directory contents in long format, showing permissions, number of links, owner, group, file size, modification date and time, and filename for each entry.

```
admin@ubuntu:~/Documents$ ls -l
total 28
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A10.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A1.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A2.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A3.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A4.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A5.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A6.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A7.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A8.txt
-rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A9.txt
-rw-rw-r-- 1 admin admin   180 Apr  4 08:57 alpha2.tar.gz
-rw-rw-r-- 1 admin admin 10240 Mar 28 09:50 alpha.tar
-rw-rw-r-- 1 admin admin  1424 Apr  4 09:25 lab.zip
-rw-rw-r-- 1 admin admin  1660 Mar 28 09:20 longfile.txt.xz
-rw-rw-r-- 1 admin admin     0 Apr  4 09:38 randomtxt.txt
-rwxrwxr-x 1 admin admin    20 Apr  4 09:37 script.sh
```

Command 7: `ls -l -S`

Syntax	`ls -l -S [path]`
Description	Lists files in long format sorted by file size, largest first. Combines the detailed view (-l) with size sorting (-S). Useful for identifying large files quickly.

```
admin@ubuntu:~/Documents$ ls -l -s
total 28
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A10.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A1.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A2.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A3.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A4.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A5.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A6.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A7.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A8.txt
 0 -rw-rw-r-- 1 admin admin     0 Mar 28 09:49 A9.txt
 4 -rw-rw-r-- 1 admin admin   180 Apr  4 08:57 alpha2.tar.gz
12 -rw-rw-r-- 1 admin admin 10240 Mar 28 09:50 alpha.tar
 4 -rw-rw-r-- 1 admin admin  1424 Apr  4 09:25 lab.zip
 4 -rw-rw-r-- 1 admin admin  1660 Mar 28 09:20 longfile.txt.xz
 0 -rw-rw-r-- 1 admin admin     0 Apr  4 09:38 randomtxt.txt
 4 -rwxrwxr-x 1 admin admin    20 Apr  4 09:37 script.sh
```

Command 8: `mkdir`

Syntax	`mkdir [directory_name]`
Description	Creates a new directory (folder) in the current location. You can create multiple directories at once. It does not create parent directories unless -p flag is used.

```
admin@ubuntu:~/Documents$ mkdir report.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          randomtxt.txt  script.sh
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  report.txt
admin@ubuntu:~/Documents$
```

Command 9: `mkdir "My Folder"`

Syntax	`mkdir "My Folder"`
Description	Creates a directory with spaces in its name. Quotes are required when directory names contain spaces; otherwise the shell interprets each word as a separate argument.

```
admin@ubuntu:~/Documents$ mkdir "My report.txt"
admin@ubuntu:~/Documents$ ls
 A10.txt   A2.txt   A4.txt   A6.txt   A8.txt   alpha2.tar.gz   lab.zip          'My report.txt'   report.txt
 A1.txt    A3.txt   A5.txt   A7.txt   A9.txt   alpha.tar       longfile.txt.xz   randomtxt.txt    script.sh
admin@ubuntu:~/Documents$
```

Command 10: `mkdir A1..10.txt`

Syntax	`mkdir A{1..10}.txt`
Description	Creates multiple directories at once using brace expansion. For example, mkdir A{1..5} creates A1, A2, A3, A4, and A5 directories simultaneously.

```
admin@ubuntu:~/Documents$ mkdir A{1..10}.txt
mkdir: cannot create directory ‘A1.txt’: File exists
mkdir: cannot create directory ‘A2.txt’: File exists
mkdir: cannot create directory ‘A3.txt’: File exists
mkdir: cannot create directory ‘A4.txt’: File exists
mkdir: cannot create directory ‘A5.txt’: File exists
mkdir: cannot create directory ‘A6.txt’: File exists
mkdir: cannot create directory ‘A7.txt’: File exists
mkdir: cannot create directory ‘A8.txt’: File exists
mkdir: cannot create directory ‘A9.txt’: File exists
mkdir: cannot create directory ‘A10.txt’: File exists
admin@ubuntu:~/Documents$
```

Command 11: `rmdir`

Syntax	`rmdir [directory_name]`

Description	Removes an empty directory. Unlike rm, rmdir only works on directories that contain no files. If the directory is not empty, it returns an error.

```
admin@ubuntu:~/Documents$ rmdir 'My report.txt'
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          randomtxt.txt  script.sh
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  report.txt
admin@ubuntu:~/Documents$
```

Command 12: `rm -rf`

Syntax	`rm -rf [directory_name]`
Description	Forcefully removes a directory and all its contents recursively. -r means recursive (all subdirectories), -f means force (no prompts). Use with extreme caution — files are permanently deleted.

```
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          randomtxt.txt  script.sh
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  report.txt
admin@ubuntu:~/Documents$ rm -rf report.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          randomtxt.txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  script.sh
```

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Command 13: `rmdir A1..10.txt`

Syntax	`rmdir A{1..10}.txt`
Description	Removes multiple empty directories at once using brace expansion. Only works if all specified directories exist and are empty.

```
admin@ubuntu:~/Documents$ rmdir A{1..10}.txt
rmdir: failed to remove 'A1.txt': Not a directory
rmdir: failed to remove 'A2.txt': Not a directory
rmdir: failed to remove 'A3.txt': Not a directory
rmdir: failed to remove 'A4.txt': Not a directory
rmdir: failed to remove 'A5.txt': Not a directory
rmdir: failed to remove 'A6.txt': Not a directory
rmdir: failed to remove 'A7.txt': Not a directory
rmdir: failed to remove 'A8.txt': Not a directory
rmdir: failed to remove 'A9.txt': Not a directory
rmdir: failed to remove 'A10.txt': Not a directory
admin@ubuntu:~/Documents$
```

Command 14: `rm`

Syntax	`rm [filename]`
Description	Removes (deletes) files. Unlike graphical deletion, rm does not move files to trash — they are permanently deleted. Use -i flag for interactive confirmation prompt.

```
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          myfolder.txt   script.sh
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt
admin@ubuntu:~/Documents$ rm myfolder.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          randomtxt.txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  script.sh
```

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SECTION 2: File Creation, Editing & Management

Command 15: `touch`

Syntax	`touch [filename]`
Description	Creates a new empty file or updates the access/modification timestamps of an existing file. Commonly used to create placeholder files quickly.

```
admin@ubuntu:~/Documents$ touch myfolder.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          myfolder.txt   script.sh
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt
admin@ubuntu:~/Documents$
```

Command 16: `touch /home/user/Music/file1.txt`

Syntax	`touch /home/user/Music/file1.txt`
Description	Creates a new empty file at the specified absolute path. If the file already exists, it updates its timestamps. Useful for creating files in directories other than the current one.

```
admin@ubuntu:~/Documents$ touch /Music /report.txt
touch: cannot touch '/Music': Permission denied
touch: cannot touch '/report.txt': Permission denied
admin@ubuntu:~/Documents$
```

Command 17: `truncate -s 10M file1.txt`

Syntax	`truncate -s [size] [filename]`
Description	Sets a file to a specific size. If the file is shorter, it is extended with null bytes. If longer, it is truncated. Useful for creating test files of exact sizes (e.g., 10M = 10 megabytes).

```
admin@ubuntu:~/Documents$ pwd
/home/admin/Documents
admin@ubuntu:~/Documents$ cd Music
admin@ubuntu:~/Documents/Music$ truncate -s 10M test.txt
admin@ubuntu:~/Documents/Music$ ls
test.txt
admin@ubuntu:~/Documents/Music$
```

Command 18: `nano`

Syntax	`nano [filename]`

Description	Opens the nano text editor in the terminal — a simple, beginner-friendly CLI text editor. Use Ctrl+O to save, Ctrl+X to exit. Ideal for quick file edits directly in the terminal.

```
admin@ubuntu:~/Documents$ cd Music
admin@ubuntu:~/Documents/Music$ truncate -s 10M test.txt
admin@ubuntu:~/Documents/Music$ ls
test.txt
admin@ubuntu:~/Documents/Music$ nano test.txt
admin@ubuntu:~/Documents/Music$
```

Command 19: gedit

Syntax	`gedit [filename]`
Description	Opens the GNOME graphical text editor. It provides a GUI-based editing experience with syntax highlighting and plugins. Requires a desktop environment to run.

```
admin@ubuntu:~/Documents/Music$ gedit test.txt
Command 'gedit' not found, but can be installed with:
sudo snap install gedit  # version 48.1, or
sudo apt  install gedit  # version 46.1-3
See 'snap info gedit' for additional versions.
admin@ubuntu:~/Documents/Music$
```

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Command 20: cp

Syntax	`cp [source] [destination]`
Description	Copies files or directories from source to destination. Use -r to copy directories recursively. Original file is preserved and a duplicate is created at the destination path.

```
admin@ubuntu:~/Documents$ cd Music
admin@ubuntu:~/Documents/Music$ cp test.txt Documents
admin@ubuntu:~/Documents/Music$ cd Documents
```

Command 21: mv

Syntax	`mv [source] [destination]`
Description	Moves or renames files and directories. If source and destination are in the same directory, it renames the file. If different, it moves the file to the new location.

```
admin@ubuntu:~/Documents/Music$ touch test2.txt
admin@ubuntu:~/Documents/Music$ mv test2.txt test.txt
admin@ubuntu:~/Documents/Music$ ls
Documents  exist  test.txt
admin@ubuntu:~/Documents/Music$
```

Command 22: `cat`

Syntax	`cat [filename]`
Description	Concatenates and displays file contents to standard output. Can display multiple files, combine them (cat f1 f2 > combined), or create files using input redirection.

```
admin@ubuntu:~/Documents/Music$ nano test2.txt
admin@ubuntu:~/Documents/Music$ cat test2.txt
hi i am sazib
admin@ubuntu:~/Documents/Music$
```

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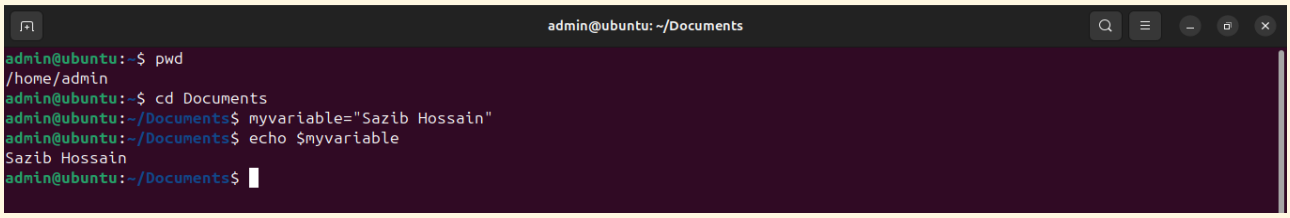
SECTION 3: Text Processing & System Information

Command 23: echo

Syntax	`echo [text or variable]`
Description	Prints text or variable values to the terminal (standard output). Widely used in scripts to display messages or variable contents. Supports escape sequences with -e flag.



Command 24: head

Syntax	`head -n [num] [filename]`
Description	Displays the first N lines of a file (default: 10). Useful for quickly previewing the beginning of large text files or log files without opening the whole file.

Command 25: tail

Syntax	`tail -n [num] [filename]`
Description	Displays the last N lines of a file (default: 10). Especially useful for monitoring log files. The -f flag follows the file in real-time as new lines are appended.

Command 26: ping

Syntax	`ping [hostname or IP]`

Description	Sends ICMP echo request packets to a network host to check connectivity and measure round-trip time. Used to diagnose network issues. Press Ctrl+C to stop.

```
admin@ubuntu:~/Documents/Music$ ping www.facebook.com
PING star-mini.c10r.facebook.com (31.13.64.35) 56(84) bytes of data.
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=1 ttl=52 time=60.7 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=2 ttl=52 time=25.0 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=3 ttl=52 time=33.2 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=4 ttl=52 time=21.6 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=5 ttl=52 time=17.9 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=6 ttl=52 time=18.5 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=7 ttl=52 time=18.0 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=8 ttl=52 time=18.6 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=9 ttl=52 time=16.5 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=10 ttl=52 time=20.2 ms
64 bytes from edge-star-mini-shv-02-ccu1.facebook.com (31.13.64.35): icmp_seq=11 ttl=52 time=18.1 ms
```

Command 27: `ipconfig`

Syntax	`ifconfig (or ip addr on modern Linux)`
Description	Displays network interface configuration details including IP address, MAC address, subnet mask, and more. On modern systems, ip addr or ip a is the preferred alternative.

```
admin@ubuntu:~/Documents/Music$ iconfig
Command 'iconfig' not found, but can be installed with:
sudo apt install ipmiutil
admin@ubuntu:~/Documents/Music$ iconfig eth0 up
Command 'iconfig' not found, but can be installed with:
sudo apt install ipmiutil
admin@ubuntu:~/Documents/Music$ ipconfig eth0 up
```

Command 28: `locate`

Syntax	`locate [filename_pattern]`
Description	Quickly searches for files by name using a pre-built database. Much faster than find but may show stale results. Update the database with sudo updatedb.

```
Try: sudo apt install <deb name>
admin@ubuntu:~/Documents/Music$ locate passwd
Command 'locate' not found, but can be installed with:
sudo apt install plocate
admin@ubuntu:~/Documents/Music$
```

Command 29: `file`

Syntax	`file [filename]`

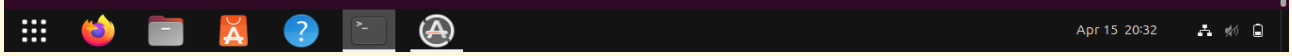
Description	Determines and displays the type of a file by examining its content (magic bytes), not just its extension. Identifies text files, executables, images, compressed files, etc.

```
admin@ubuntu:~/Documents/Music$ file test2.txt
test2.txt: ASCII text
admin@ubuntu:~/Documents/Music$
```

Command 30: `chmod`

Syntax	`chmod [permissions] [filename]`
Description	Changes file or directory permissions. Permissions control read (r), write (w), and execute (x) access for owner, group, and others. Can use symbolic (u+x) or numeric (755) mode.

```
test: cannot open `test' (No such file or directory)
admin@ubuntu:~/Documents/Music$ file test2.txt
test2.txt: ASCII text
admin@ubuntu:~/Documents/Music$ chmod 755 test2.txt
admin@ubuntu:~/Documents/Music$
```



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Command 31: `chown`

Syntax	`chown [owner]:[group] [filename]`
Description	Changes the ownership of a file or directory. Requires superuser (sudo) privileges. Useful for transferring file ownership between users or setting service-specific ownership.

```
chown: invalid user: 'user1:group'
admin@ubuntu:~/Documents/Music$ chown admin:group test2.txt
chown: invalid group: 'admin:group'
admin@ubuntu:~/Documents/Music$
```

Command 32: `grep`

Syntax	`grep [pattern] [filename]`
Description	Searches for lines matching a pattern (regular expression or string) in files and prints matching lines. Extremely powerful for searching log files, source code, or configuration files.

```
sudo apt install tpmtutil
admin@ubuntu:~/Documents/Music$ grep "sazib" test2.txt
hi i am sazib
admin@ubuntu:~/Documents/Music$
```

Command 33: `find`

Syntax	`find [path] -name [pattern]`
Description	Recursively searches the directory tree for files and directories matching given criteria (name, type, size, modification time, permissions, etc.). Much more flexible than locate.

```
admin@ubuntu:~/Documents/Music$ find /home -name "*.txt"
/home/admin/snap/firefox/common/.mozilla/firefox/i96szirp.default/pkcs11.txt
/home/admin/.cache/tracker3/files/last-crawl.txt
/home/admin/.cache/tracker3/files/first-index.txt
/home/admin/A1.txt
/home/admin/longfile21.txt
/home/admin/A3.txt
/home/admin/A2.txt
/home/admin/A4.txt
/home/admin/A6.txt
/home/admin/A10.txt
```

Command 34: `clear`

Syntax	`clear`
Description	Clears all visible text from the terminal screen. The scroll history is usually preserved (you can scroll up). Equivalent to pressing Ctrl+L in most terminals.

```
admin@ubuntu: ~/Documents/Music
admin@ubuntu:~/Documents/Music$ clear
```

Command 35: `history`

Syntax	`history [n]`
Description	Displays the list of previously executed commands with their line numbers. Useful for recalling and re-running past commands. History is stored in ~/.bash_history.

```
admin@ubuntu: ~/Documents/Music
admin@ubuntu:~/Documents/Music$ history 5
  333  find /Documents/Music -name "*.txt"
  334  find /home -name "*.txt"
  335  clear
  336  clear
  337  history 5
admin@ubuntu:~/Documents/Music$
```

SECTION 4: Important System Directories & Files

Command 36: /boot/vmlinuz

Syntax	`ls -lh /boot/vmlinuz*`
Description	The compressed Linux kernel image stored in /boot. vmlinuz is the bootable, compressed kernel that the bootloader (e.g., GRUB) loads into memory during system startup.

```
admin@ubuntu: ~
admin@ubuntu:~$ ls -l /boot/vmlinuz
lrwxrwxrwx 1 root root 25 Jan 25 16:29 /boot/vmlinuz -> vmlinuz-6.14.0-37-generic
admin@ubuntu:~$
```

Command 37: /dev/null

Syntax	`echo 'text' > /dev/null`
Description	/dev/null is a special device file (null device) that discards all data written to it. Reading from it returns EOF immediately. Commonly used to suppress command output.

```
admin@ubuntu: ~
admin@ubuntu:~$ ls -l /boot/vmlinuz
lrwxrwxrwx 1 root root 25 Jan 25 16:29 /boot/vmlinuz -> vmlinuz-6.14.0-37-generic
admin@ubuntu:~$ echo "this disappears" > /dev/null
admin@ubuntu:~$
```

Command 38: /etc/exports

Syntax	`cat /etc/exports`
Description	Configuration file for NFS (Network File System) exports. Lists directories to be shared with remote systems over the network, along with access permissions and client specifications.

```
admin@ubuntu:~$ cat /etc/exports
cat: /etc/exports: No such file or directory
admin@ubuntu:~$
```

Command 39: /etc/init.d

Syntax	`ls /etc/init.d`

Description	Directory containing SysV-style init scripts for starting, stopping, and managing system services. On systemd-based systems (Ubuntu 16+), these are largely replaced by systemctl unit files.

```
admin@ubuntu:~$ ls /etc/init.d
alsa-utils  bluetooth         dbus         keyboard-setup.sh  plymouth-log  speech-dispatcher  sysstat              whoopsie
anacron     console-setup.sh  gdm3         kmod               procps        spice-vdagent      ufw                  x11-common
apparmor    cron              grub-common  openvpn            rsync         ssh                unattended-upgrades
apport      cups              kerneloops   plymouth           saned         sssd               uuidd
admin@ubuntu:~$
```

Command 40: /etc/hosts

Syntax	`cat /etc/hosts`
Description	A plain-text file mapping hostnames to IP addresses locally, before DNS is consulted. Used for local hostname resolution, blocking websites, or defining custom network aliases.

```
admin@ubuntu:~$ cat /etc/hosts
127.0.0.1       localhost.localdomain   localhost
::1             localhost6.localdomain6 localhost6

# The following lines are desirable for IPv6 capable hosts
::1     localhost ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
admin@ubuntu:~$
```

Command 41: /etc/passwd

Syntax	`cat /etc/passwd`
Description	Contains one entry per user account on the system with fields: username, password placeholder, UID, GID, GECOS info, home directory, and login shell. Passwords are stored in /etc/shadow.

```
admin@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
```

Command 42: /etc/profile

Syntax	`cat /etc/profile`

Description	System-wide shell startup script executed for login shells. Sets environment variables (PATH, EDITOR, etc.) and runs scripts in /etc/profile.d/ for all users upon login.

```
admin@ubuntu:~$ cat /etc/profile
# /etc/profile: system-wide .profile file for the Bourne shell (sh(1))
# and Bourne compatible shells (bash(1), ksh(1), ash(1), ...).

if [ "${PS1-}" ]; then
  if [ "${BASH-}" ] && [ "$BASH" != "/bin/sh" ]; then
    # The file bash.bashrc already sets the default PS1.
    # PS1='\h:\w\$ '
    if [ -f /etc/bash.bashrc ]; then
      . /etc/bash.bashrc
```

Command 43: /etc/securetty

Syntax	`cat /etc/securetty`
Description	Lists terminal devices (ttys) from which root can log in directly. If a terminal is not listed here, root login from that terminal is denied, enhancing system security.

```
admin@ubuntu:~$ cat /etc/securetty
cat: /etc/securetty: No such file or directory
admin@ubuntu:~$ cat /etc/securettys
cat: /etc/securettys: No such file or directory
admin@ubuntu:~$ file /etc/securetty
/etc/securetty: cannot open `/etc/securetty' (No such file or directory)
```

Command 44: /usr/bin

Syntax	`ls /usr/bin`
Description	Directory containing most user-accessible executable programs and commands. Binaries here are available to all users. Examples: ls, grep, find, python3, git, etc.

```
admin@ubuntu:~$ ls /usr/bin
'['                                 gnome-control-center        mtr-packet                  spa-inspect
 aa-enabled                         gnome-disk-image-mounter    mv                          spa-json-dump
 aa-exec                            gnome-disks                 namei                       spa-monitor
 aa-features-abi                    gnome-extensions            nano                        spa-resample
 aconnect                           gnome-extensions-app        nautilus                    spd-conf
 acpidbg                            gnome-font-viewer           nautilus-autorun-software   spd-say
 add-apt-repository                 gnome-help                  nautilus-sendto             spdsend
 addpart                            gnome-keyring               nawk                        speaker-test
 airscan-discover                   gnome-keyring-3             nc                          speech-dispatcher
```

Command 45: /usr/include

Syntax	`ls /usr/include`

Description	Contains C and C++ header files (.h files) used when compiling programs. System libraries store their API header files here for use by developers and build systems.

```
admin@ubuntu:~$ ls /usr/include
aio.h        drm                fts.h          libintl.h    netatalk     poll.h          sched.h        stdlib.h       ucontext.h
aliases.h    elf.h              ftw.h          limits.h     netax25      printf.h        scsi           string.h       ulimit.h
alloca.h     endian.h           gconv.h        link.h       netdb.h      proc_service.h  search.h       strings.h      unistd.h
argp.h       envz.h             gdb            linux        neteconet    protocols       semaphore.h    sudo_plugin.h  utime.h
argz.h       err.h              getopt.h       locale.h     netinet      pthread.h       setjmp.h       syscall.h      utmp.h
ar.h         errno.h            glob.h         malloc.h     netipx       pty.h           sgtty.h        sysexits.h     utmpx.h
arpa         error.h            gnu-versions.h math.h       netiucv      pwd.h           shadow.h       syslog.h       values.h
asm-generic  execinfo.h         grp.h          mcheck.h     netpacket    python3.12      signal.h       tar.h          video
assert.h     fcntl.h            gshadow.h      memory.h     netrom       rdma            sound          termio.h       wait.h
byteswap.h   features.h         iconv.h        misc         netrose      re_comp.h       spawn.h        termios.h      wchar.h
```

Command 46: /usr/share

Syntax	`ls /usr/share`
Description	Contains architecture-independent data shared across packages: documentation, icons, man pages, locale data, application data files, and other static shared resources.

```
admin@ubuntu:~$ ls /usr/share
accountsservice        gcc                      libc-bin                      pyshared
aclocal                GConf                    libchewing                    python3
alsa                   gdb                      libdebuginfod-common          python-apt
alsa-base              gdm                      libdrm                        python-babel-localedata
alsa-card-profile      gettext                  libgcrypt20                   readline
applications           ghostscript              libgedit-gtksourceview-300    rsync
apport                 git-core                 libgnomekbd                   rsyslog
appstream              gitweb                   libgweather-4                 rygel
```

Command 47: /usr/lib

Syntax	`ls /usr/lib`
Description	Contains shared library files (.so files) and internal program data used by executables in /usr/bin and /usr/sbin. These libraries provide reusable code for multiple programs.

```
admin@ubuntu:~$ ls /usr/lib
apg                    file                     klibc                                nvidia                 speech-dispatcher-modules
apparmor               firewalld                klibc-sw0VayLfV0hmLGCQE9vyfOnj81g.so openssh               ssl
apt                    firmware                 linux                                openvpn                sysctl.d
aspell                 girepository-1.0         linux-boot-probes                    os-probes              sysstat
binfmt.d               git-core                 linux-hwe-6.14-tools-6.14.0-37       os-release             systemd
brltty                 gnome-session            linux-sound-base                     pam.d                  sysusers.d
cloud-init             gnome-settings-daemon-3.0 linux-tools                         pcmciautils            tmpfiles.d
cnf-update-db          gnome-settings-daemon-46 llvm-18                              pcrlock.d              ubiquity
command-not-found      gnome-shell              locale                               pm-utils               ubuntu-advantage
console-setup          gnupg                    lsb                                  policykit-1            ubuntu-release-upgrader
```

Command 48: /proc/cpuinfo

Syntax	`cat /proc/cpuinfo`
Description	A virtual file (in the proc filesystem) showing detailed CPU information: processor model, vendor, speed (MHz), cache size, flags indicating supported instruction sets, and core count.

```
admin@ubuntu:~$ cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 142
model name      : Intel(R) Core(TM) i7-8665U CPU @ 1.90GHz
stepping        : 12
microcode       : 0xffffffff
cpu MHz         : 2112.004
cache size      : 8192 KB
physical id     : 0
siblings        : 2
core id         : 0
cpu cores       : 2
```

Command 49: `/proc/interrupts`

Syntax	`cat /proc/interrupts`
Description	A virtual file listing hardware interrupt (IRQ) statistics for each CPU. Shows how many times each device (keyboard, disk, network) has triggered an interrupt per CPU core.

```
admin@ubuntu:~$ cat /proc/interrupts
           CPU0       CPU1
  0:        204          0  IO-APIC   2-edge      timer
  1:          0        612  IO-APIC   1-edge      i8042
  8:          0          0  IO-APIC   8-edge      rtc0
  9:          0          0  IO-APIC   9-fasteoi   acpi
 12:       1930          0  IO-APIC  12-edge      i8042
 14:       2360          0  IO-APIC  14-edge      ata_piix
 15:          0          0  IO-APIC  15-edge      ata_piix
 18:          0          4  IO-APIC  18-fasteoi   vmwgfx
 19:          0       1707  IO-APIC  19-fasteoi   ehci_hcd:usb2, enp0s3
 20:       3513          0  IO-APIC  20-fasteoi   vboxguest
```

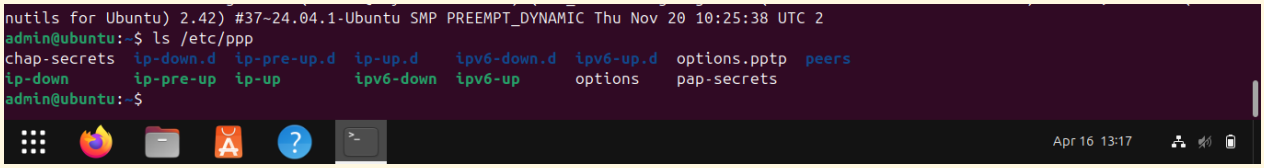
Command 50: `/version`

Syntax	`cat /proc/version`
Description	Displays the Linux kernel version string including the kernel version number, the GCC compiler version used to compile it, and the build timestamp.

```
PIW:          0          0   Posted-interrupt wakeup event
admin@ubuntu:~$ cat /proc/version
Linux version 6.14.0-37-generic (buildd@lcy02-amd64-031) (x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, GNU ld (GNU Bi
nutils for Ubuntu) 2.42) #37~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Nov 20 10:25:38 UTC 2
admin@ubuntu:~$
```

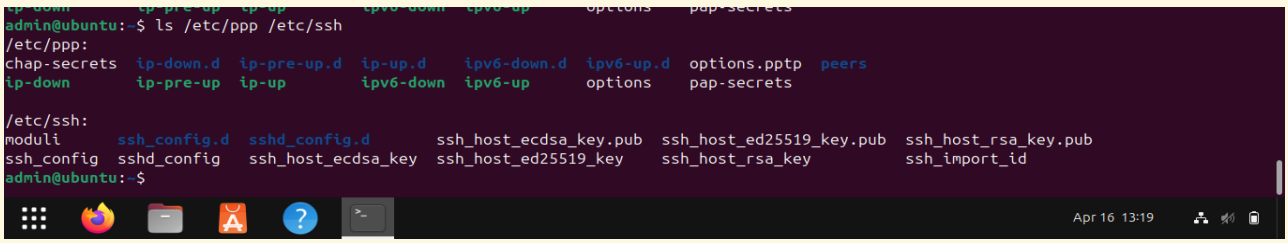
Command 51: `ls /etc/ppp`

Syntax	`ls /etc/ppp`
Description	Lists the contents of /etc/ppp, the configuration directory for PPP (Point-to-Point Protocol) connections — used for dial-up and some VPN/serial link configurations.



Command 52: `ls /etc/ppp /etc/ssh`

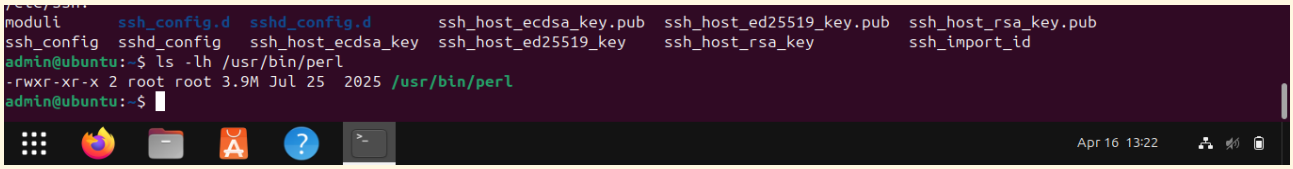
Syntax	`ls /etc/ppp /etc/ssh`
Description	Lists contents of two directories in one command. Both PPP and SSH configuration directories are displayed sequentially, each preceded by its path as a label.



SECTION 5: Date, History, Variables & Shell Features

Command 53: `ls -lh /usr/bin/perl`

Syntax	`ls -lh /usr/bin/perl`
Description	Lists details of the Perl interpreter binary with human-readable file size (-h). Shows the path, permissions, owner, and size of the installed Perl binary.



Command 54: `date`

Syntax	`date [+format]`
Description	Displays the current system date and time. Can be formatted using format specifiers (e.g., date '+%Y-%m-%d'). With sudo, can be used to set the system date and time.

Command 55: `cal 5 2030`

Syntax	`cal [month] [year]`
Description	Displays a calendar for the specified month and year. Without arguments, shows the current month. A useful utility for checking dates and planning schedules from the terminal.

Command 56: `!3`

Syntax	`![history_number]`
Description	Re-executes the command at line number 3 in the bash history. The ! (bang) operator followed by a number replays a specific command from the history list without retyping it.

Command 57: `history 3`

Syntax	`history [n]`
Description	Displays the last N commands in the history list. Here it shows the last 3 commands. Useful for quickly reviewing recent activity without listing the entire history.

Command 58: `!-3`

Syntax	`!-[n]`
Description	Re-executes the command that was 3 commands ago (third from last). Negative offsets count back from the current command. Convenient for repeating recent commands.

Command 59: `!ls`

Syntax	`![command_prefix]`

Description	Re-executes the most recent command that starts with 'ls'. The ! followed by a string searches backward in history for the most recent matching command and reruns it.

Command 60: `Local Variable`

Syntax	`variable_name=value  (no spaces around =)`
Description	A variable defined in the shell that is only accessible within the current shell session. Not available to child processes. To make it available to child processes, use export.

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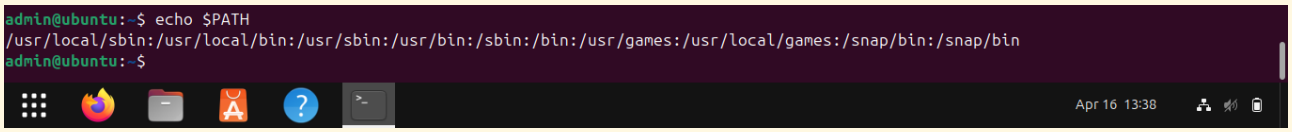
Command 61: `echo $HISTSIZE`

Syntax	`echo $HISTSIZE`
Description	Displays the value of the HISTSIZE environment variable, which defines how many commands bash stores in memory during a session. Typically set to 1000 by default.

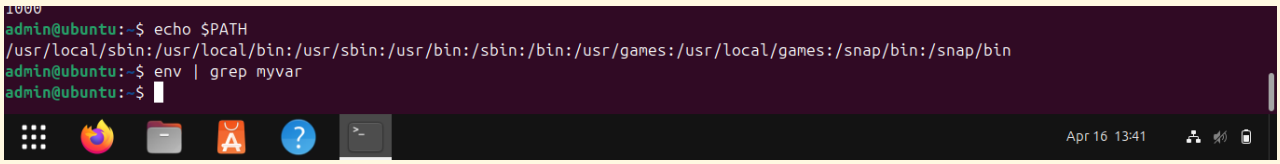
Command 62: `echo $PATH`

Syntax	`echo $PATH`
Description	Displays the PATH environment variable — a colon-separated list of directories the shell searches when you type a command. The shell looks in each directory left-to-right for the executable.



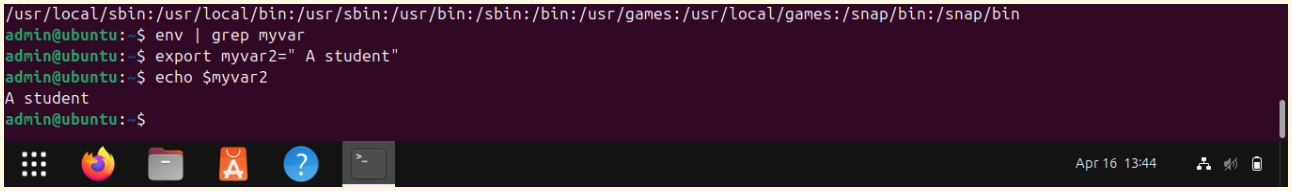
Command 63: `env | grep variable1`

Syntax	`env | grep [variable_name]`
Description	Lists all environment variables (env) and pipes (|) the output to grep, which filters for lines containing 'variable1'. Used to check if a specific variable is set and exported.



Command 64: `export variable1`

Syntax	`export [variable_name[=value]]`
Description	Marks a shell variable for export to child processes (subshells, scripts, programs). Without export, variables are local to the current shell and inaccessible to processes it spawns.



Command 65: `type cd`

Syntax	`type [command_name]`
Description	Reveals how the shell would interpret a command name — whether it is a shell builtin, alias, function, or external executable. Helps understand command precedence.

Command 66: `which ls`

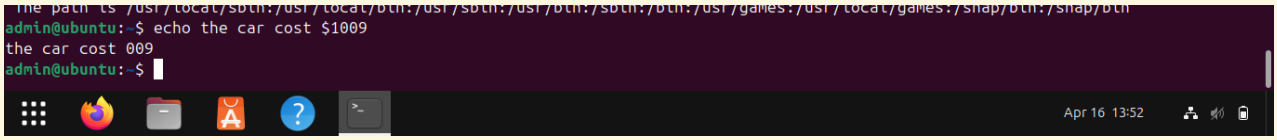
Syntax	`which [command_name]`
Description	Locates the full path of an executable by searching directories listed in PATH. Useful for identifying which version of a command will be run when multiple versions are installed.

Command 67: `echo "The path is $PATH"`

Syntax	`echo "text with $VAR"`
Description	Double quotes allow variable expansion inside. The $PATH variable is replaced with its actual value in the output. Use double quotes when you want variables to be interpreted.

Command 68: `echo The car costs $100`

Syntax	`echo The car costs $100`
Description	Without quotes, $1 is interpreted as a variable (which is usually empty or a positional parameter). The output may be unexpected — use single quotes to prevent variable expansion.



Command 69: `echo 'The car costs $100'`

Syntax	`echo 'literal $string'`
Description	Single quotes prevent ALL special character interpretation and variable expansion. Everything inside single quotes is treated as a literal string, exactly as typed.

Command 70: `echo Today is date`

Syntax	`echo Today is date`
Description	Prints the literal text 'Today is date'. The word 'date' is treated as plain text (not the date command) unless enclosed in backticks or $() command substitution.

Command 71: `echo Today is 'date'`

Syntax	`echo Today is $(date)`
Description	To get the output of the date command inline, use command substitution: $(date) or backticks. Single-quoted 'date' would print literally; $(date) executes the command.

Command 72: `cal 1 2030; cal 2 2030; cal 3 2030`

Syntax	`command1; command2; command3`
Description	Runs multiple commands sequentially separated by semicolons. Each command runs after the previous one completes, regardless of success or failure. Here it shows Jan, Feb, Mar 2030 calendars.

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Command 73: `echo D*`

Syntax	`echo D*`
Description	The * is a wildcard that expands to all files/directories starting with 'D' in the current directory. The shell performs glob expansion before passing arguments to echo.

Command 74: `ls /etc/ppp || echo failed`

Syntax	`command1 || command2`
Description	OR operator: runs command2 ONLY if command1 fails (exits with non-zero status). If ls /etc/ppp succeeds, 'failed' is not printed. Useful for fallback or error-handling in scripts.

```
admin@ubuntu:~$ ls /etc/ppp || echo filed
chap-secrets  ip-down.d  ip-pre-up.d  ip-up.d    ipv6-down.d  ipv6-up.d  options.pptp  peers
ip-down       ip-pre-up  ip-up        ipv6-down  ipv6-up      options    pap-secrets
admin@ubuntu:~$
```

Command 75: `ls /etc/ppp && echo success`

Syntax	`command1 && command2`
Description	AND operator: runs command2 ONLY if command1 succeeds (exits with zero status). If ls /etc/ppp fails, 'success' is not printed. Used for conditional execution chains.

```
admin@ubuntu:~$ ls /etc/ppp && echo filed
chap-secrets  ip-down.d  ip-pre-up.d  ip-up.d    ipv6-down.d  ipv6-up.d  options.pptp  peers
ip-down       ip-pre-up  ip-up        ipv6-down  ipv6-up      options    pap-secrets
filed
admin@ubuntu:~$
```

Apr 16 14:08

Command 76: `man ls`

Syntax	`man [command_name]`
Description	Opens the manual (man) page for a command, providing comprehensive documentation including syntax, options, examples, and related commands. Press q to exit. Use / to search within the page.

```
admin@ubuntu: ~
LS(1)                                User Commands                                LS(1)

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List  information  about the FILEs (the current directory by default).  Sort entries alphabetically if none of -cftuvSUX nor
       --sort is specified.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
```

SECTION 6: Help, Documentation & Info Commands

Command 77: `man -k copy`

Syntax	`man -k [keyword]`
Description	Searches all man page names and short descriptions for the given keyword and lists matching entries. Equivalent to the apropos command. Useful for discovering unknown commands.

Command 78: `whatis ls`

Syntax	`whatis [command_name]`
Description	Displays the one-line description from the man page for the specified command. Provides a quick summary of what a command does without opening the full manual page.

Command 79: `whereis ls`

Syntax	`whereis [command_name]`
Description	Locates the binary, source code, and man page for a command. Unlike which, it searches standard system locations and returns paths for all three components (binary, source, manual).

Command 80: `info ls`

Syntax	`info [command_name]`
Description	Displays the GNU info documentation for a command — often more detailed than man pages. Structured in a hyperlinked node format. Press q to exit, use arrow keys to navigate.

```
options over the years.  They are described in the subsections below;
within each section, options are listed alphabetically (ignoring case).
The division of options into the subsections is not absolute, since some
options affect more than one aspect of 'ls''s operation.

-----Info: (coreutils)ls invocation, 56 lines --Top-----------------------------------------------------------------------
Welcome to Info version 7.1.  Type H for help, h for tutorial.
```

Command 81: `cat --help`

Syntax	`[command] --help`
Description	Displays a concise usage summary and available options for a command. The --help flag is supported by most GNU/Linux commands and is quicker than reading the full man page.

```
admin@ubuntu:~$ cat --help
Usage: cat [OPTION]... [FILE]...
Concatenate FILE(s) to standard output.

With no FILE, or when FILE is -, read standard input.

  -A, --show-all           equivalent to -vET
  -b, --number-nonblank    number nonempty output lines, overrides -n
  -e                       equivalent to -vE
  -E, --show-ends          display $ at end of each line
```

Command 82: `gzip`

Syntax	`gzip [filename]`
Description	Compresses a file using the GNU zip compression algorithm (.gz format). The original file is replaced by the compressed version with .gz appended. Use gunzip to decompress.

```
admin@ubuntu:~/Documents$ touch report.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
admin@ubuntu:~/Documents$ gzip report.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt.gz  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
admin@ubuntu:~/Documents$
```

SECTION 7: Compression & Archiving (gzip, tar, zip)

Command 83: `gzip -l longfile.txt.gz`

Syntax	`gzip -l [file.gz]`
Description	Lists compression information about a .gz file without decompressing it: compressed size, uncompressed size, compression ratio, and original filename.

```
admin@ubuntu:~/Documents$ touch report.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
admin@ubuntu:~/Documents$ gzip report.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt.gz  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
admin@ubuntu:~/Documents$ gzip -l report.txt.gz
         compressed        uncompressed  ratio uncompressed_name
                 31                   0   0.0% report.txt
admin@ubuntu:~/Documents$
```

Command 84: `gunzip`

Syntax	`gunzip [file.gz]`
Description	Decompresses a .gz file created by gzip. Restores the original file and removes the .gz version. Equivalent to running gzip -d.

```
admin@ubuntu:~/Documents$ gzip report.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt.gz  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
admin@ubuntu:~/Documents$ gzip -l report.txt.gz
         compressed        uncompressed  ratio uncompressed_name
                 31                   0   0.0% report.txt
admin@ubuntu:~/Documents$ gunzip report.txt.gz
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
```

Command 85: `bzip2`

Syntax	`bzip2 [filename]`
Description	Compresses files using the bzip2 algorithm (.bz2 format), which typically achieves better compression than gzip but is slower. The original file is replaced with the .bz2 compressed version.

```
admin@ubuntu:~/Documents$ gunzip report.txt.gz
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
admin@ubuntu:~/Documents$ bzip2 report.txt
Command 'bzip2' not found, but can be installed with:
sudo apt install bzip2
admin@ubuntu:~/Documents$
```

Command 86: bunzip2

Syntax	`bunzip2 [file.bz2]`
Description	Decompresses files compressed with bzip2. Restores the original file and removes the .bz2 archive. Equivalent to bzip2 -d.


```
admin@ubuntu:~/Documents$ gunzip report.txt.gz
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
admin@ubuntu:~/Documents$ bzip2 report.txt
Command 'bzip2' not found, but can be installed with:
sudo apt install bzip2
admin@ubuntu:~/Documents$
```


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Command 87: tar -cf

Syntax	`tar -cf archive.tar [files/dirs]`
Description	Creates a tar archive (-c) and saves it to the specified file (-f). tar itself does not compress; it bundles multiple files/directories into a single archive file.

```
admin@ubuntu:~/Documents$ touch lab{1..10}.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A3.txt  A6.txt  A9.txt         lab10.txt  lab3.txt  lab6.txt  lab9.txt         Music          script.sh
A1.txt   A4.txt  A7.txt  alpha2.tar.gz  lab1.txt   lab4.txt  lab7.txt  lab.zip          randomtxt.txt  test2,txt
A2.txt   A5.txt  A8.txt  alpha.tar      lab2.txt   lab5.txt  lab8.txt  longfile.txt.xz  report.txt
admin@ubuntu:~/Documents$ tar -cf alpha.tar lab*.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A3.txt  A6.txt  A9.txt         lab10.txt  lab3.txt  lab6.txt  lab9.txt         Music          script.sh
A1.txt   A4.txt  A7.txt  alpha2.tar.gz  lab1.txt   lab4.txt  lab7.txt  lab.zip          randomtxt.txt  test2,txt
A2.txt   A5.txt  A8.txt  alpha.tar      lab2.txt   lab5.txt  lab8.txt  longfile.txt.xz  report.txt
```

Command 88: tar -czf

Syntax	`tar -czf archive.tar.gz [files/dirs]`

Description	Creates a gzip-compressed tar archive. Combines tar bundling with gzip compression. The resulting .tar.gz (or .tgz) file is smaller than a plain tar archive.

```
A2.txt   A5.txt  A8.txt  alpha.tar      lab2.txt   lab5.txt  lab8.txt  longfile.txt.xz  report.txt
admin@ubuntu:~/Documents$ tar -cf alpha.tar lab*.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A3.txt  A6.txt  A9.txt         lab10.txt  lab3.txt  lab6.txt  lab9.txt         Music          script.sh
A1.txt   A4.txt  A7.txt  alpha2.tar.gz  lab1.txt   lab4.txt  lab7.txt  lab.zip          randomtxt.txt  test2,txt
A2.txt   A5.txt  A8.txt  alpha.tar      lab2.txt   lab5.txt  lab8.txt  longfile.txt.xz  report.txt
admin@ubuntu:~/Documents$ tar -czf alpha.tar.gz lab*.txt
admin@ubuntu:~/Documents$ ls
A10.txt  A3.txt  A6.txt  A9.txt         alpha.tar.gz  lab2.txt  lab5.txt  lab8.txt  longfile.txt.xz  report.txt
A1.txt   A4.txt  A7.txt  alpha2.tar.gz  lab10.txt     lab3.txt  lab6.txt  lab9.txt  Music            script.sh
A2.txt   A5.txt  A8.txt  alpha.tar      lab1.txt      lab4.txt  lab7.txt  lab.zip   randomtxt.txt    test2,txt
```

Command 89: `tar -cjf`

Syntax	`tar -cjf archive.tar.bz2 [files/dirs]`
Description	Creates a bzip2-compressed tar archive. Provides better compression ratio than gzip (-z) but is slower. Produces .tar.bz2 files.

```
admin@ubuntu:~/Documents$ gunzip report.txt.gz
admin@ubuntu:~/Documents$ ls
A10.txt  A2.txt  A4.txt  A6.txt  A8.txt  alpha2.tar.gz  lab.zip          Music          report.txt  test2,txt
A1.txt   A3.txt  A5.txt  A7.txt  A9.txt  alpha.tar      longfile.txt.xz  randomtxt.txt  script.sh
admin@ubuntu:~/Documents$ bzip2 report.txt
Command 'bzip2' not found, but can be installed with:
sudo apt install bzip2
admin@ubuntu:~/Documents$
```

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Command 90: `tar -tf`

Syntax	`tar -tf archive.tar`
Description	Lists the contents of a tar archive without extracting it. Shows all files and directories stored inside the archive. Add -v for verbose output with permissions and sizes.

```
A2.txt   A5.txt  A8.txt  alpha.tar      lab1.txt      lab4.txt  lab7.txt  lab.zip   randomtxt.txt    test2,txt
admin@ubuntu:~/Documents$ tar -tf alpha.tar
lab10.txt
lab1.txt
lab2.txt
lab3.txt
lab4.txt
lab5.txt
lab6.txt
lab7.txt
lab8.txt
lab9.txt
admin@ubuntu:~/Documents$
```

Command 91: `tar -tzf`

Syntax	`tar -tzf archive.tar.gz`
Description	Lists contents of a gzip-compressed tar archive without extracting. Combines -t (list), -z (gzip decompress), -f (file). Useful for inspecting .tar.gz files before extraction.

```
admin@ubuntu:~/Documents$ tar -tzf alpha.tar.gz
lab10.txt
lab1.txt
lab2.txt
lab3.txt
lab4.txt
lab5.txt
lab6.txt
lab7.txt
lab8.txt
lab9.txt
```

Command 92: `tar -xf`

Syntax	`tar -xf archive.tar`
Description	Extracts the contents of a tar archive in the current directory. Use -C [directory] to extract to a specific location. Does not handle compression automatically.

```
admin@ubuntu:~/Documents$ tar -xf alpha.tar
admin@ubuntu:~/Documents$ ls
A10.txt  A3.txt  A6.txt  A9.txt         alpha.tar.gz  lab2.txt  lab5.txt  lab8.txt  longfile.txt.xz  report.txt
A1.txt   A4.txt  A7.txt  alpha2.tar.gz  lab10.txt     lab3.txt  lab6.txt  lab9.txt  Music            script.sh
A2.txt   A5.txt  A8.txt  alpha.tar      lab1.txt      lab4.txt  lab7.txt  lab.zip   randomtxt.txt    test2,txt
admin@ubuntu:~/Documents$
```

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Command 93: `tar -xzf`

Syntax	`tar -xzf archive.tar.gz`
Description	Extracts a gzip-compressed tar archive. Combines -x (extract), -z (decompress gzip), -f (file). Most commonly used tar extraction command for .tar.gz files.

```
admin@ubuntu:~/Documents$ tar -xf alpha.tar
admin@ubuntu:~/Documents$ ls
A10.txt  A3.txt  A6.txt  A9.txt         alpha.tar.gz  lab2.txt  lab5.txt  lab8.txt  longfile.txt.xz  report.txt
A1.txt   A4.txt  A7.txt  alpha2.tar.gz  lab10.txt     lab3.txt  lab6.txt  lab9.txt  Music            script.sh
A2.txt   A5.txt  A8.txt  alpha.tar      lab1.txt      lab4.txt  lab7.txt  lab.zip   randomtxt.txt    test2,txt
admin@ubuntu:~/Documents$ tar -xzf alpha.tar.gz
admin@ubuntu:~/Documents$ ls
A10.txt  A3.txt  A6.txt  A9.txt         alpha.tar.gz  lab2.txt  lab5.txt  lab8.txt  longfile.txt.xz  report.txt
A1.txt   A4.txt  A7.txt  alpha2.tar.gz  lab10.txt     lab3.txt  lab6.txt  lab9.txt  Music            script.sh
A2.txt   A5.txt  A8.txt  alpha.tar      lab1.txt      lab4.txt  lab7.txt  lab.zip   randomtxt.txt    test2,txt
admin@ubuntu:~/Documents$
```

Command 94: `zip`

Syntax	`zip archive.zip [files]`
Description	Creates a zip archive compatible with Windows and other systems. Supports compression and can archive multiple files/directories into a single .zip file.

```
admin@ubuntu:~/Documents$ ls
A10.txt  A3.txt  A6.txt  A9.txt         alpha.tar.gz  lab2.txt  lab5.txt  lab8.txt  longfile.txt.xz  report.txt
A1.txt   A4.txt  A7.txt  alpha2.tar.gz  lab10.txt     lab3.txt  lab6.txt  lab9.txt  Music            script.sh
A2.txt   A5.txt  A8.txt  alpha.tar      lab1.txt      lab4.txt  lab7.txt  lab.zip   randomtxt.txt    test2,txt
admin@ubuntu:~/Documents$ zip beata.zip report.txt
  adding: report.txt (stored 0%)
admin@ubuntu:~/Documents$
```

Command 95: `unzip`

Syntax	`unzip archive.zip`
Description	Extracts the contents of a .zip file. Use -d [directory] to extract to a specific folder. Lists contents with -l, tests integrity with -t.

```
A2.txt   A5.txt  A8.txt  alpha.tar      lab1.txt      lab4.txt  lab7.txt  lab.zip   randomtxt.txt    test2,txt
admin@ubuntu:~/Documents$ zip beata.zip report.txt
  adding: report.txt (stored 0%)
admin@ubuntu:~/Documents$ unzip beata.zip
Archive:  beata.zip
replace report.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename:
```

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Command 96: `whoami`

Syntax	`whoami`
Description	Prints the username of the currently logged-in user (the effective user running the process). Commonly used in scripts to check privileges or personalize output.

```
replace report.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: Y
 extracting: report.txt
admin@ubuntu:~/Documents$ whoami
admin
admin@ubuntu:~/Documents$
```

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SECTION 8: User Management & Security Commands

Command 97: `sudo passwd root`

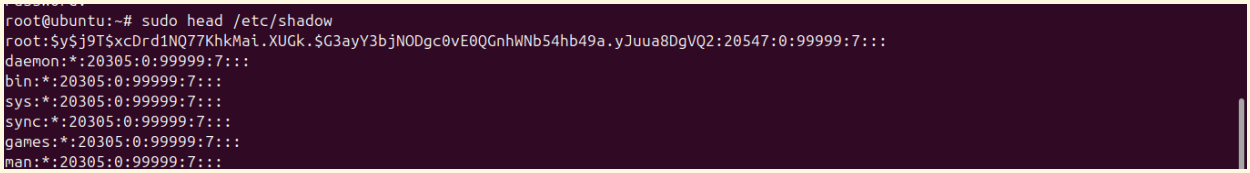
Syntax	`sudo passwd [username]`
Description	Changes the password for the specified user (here: root). Requires sudo privileges. If the root account has no password (common in Ubuntu), this command sets one, enabling direct root login.

Command 98: `su -`

Syntax	`su - [username]`
Description	Switches to another user account (defaults to root if no username given). The hyphen (-) loads the target user's environment (home directory, PATH, etc.) as a full login shell.

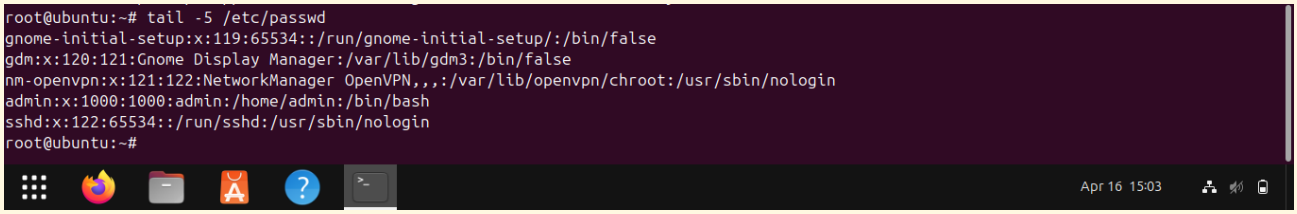
Command 99: `head /etc/shadow`

Syntax	`head [filename]`
Description	Displays the first 10 lines of /etc/shadow — the file storing hashed user passwords and password aging info. Only root can read this file. Readable only with sudo/root privileges.



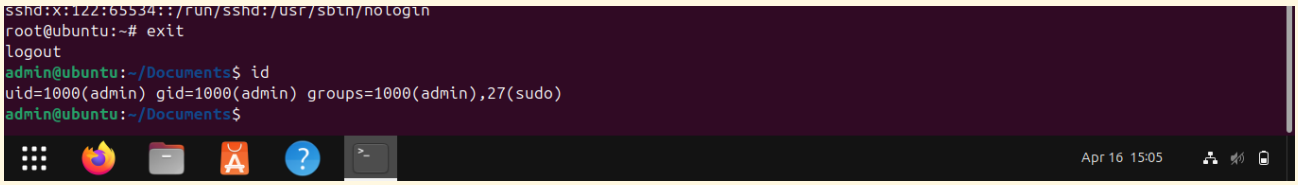
Command 100: `tail -5 /etc/passwd`

Syntax	`tail -5 /etc/passwd`
Description	Displays the last 5 lines of /etc/passwd. Useful for checking recently added user accounts, as new accounts are typically appended at the end of the file.



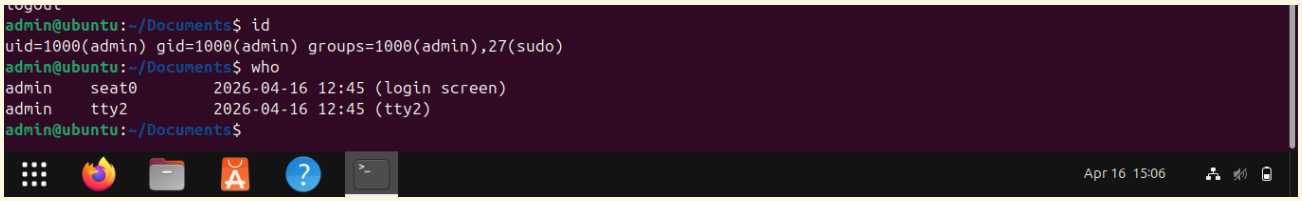
Command 101: `id`

Syntax	`id [username]`
Description	Displays the user ID (UID), primary group ID (GID), and all supplementary group memberships for the current user (or a specified user). Essential for understanding user privileges.



Command 102: `who`

Syntax	`who`
Description	Shows information about users currently logged into the system: username, terminal (tty), login time, and remote hostname or IP. Shows all active sessions.



Command 103: `last`

Syntax	`last [username]`

Description	Displays a list of recent logins, logouts, and system reboots by reading /var/log/wtmp. Shows username, terminal, source IP/hostname, login time, and duration.

```
admin@ubuntu:~/Documents$ last
admin    tty2         tty2             Thu Apr 16 12:45   still logged in
admin    seat0        login screen     Thu Apr 16 12:45   still logged in
reboot   system boot  6.14.0-37-generi Thu Apr 16 12:34   still running
admin    tty2         tty2             Thu Apr 16 06:00 - crash  (06:34)
admin    seat0        login screen     Thu Apr 16 06:00 - crash  (06:34)
reboot   system boot  6.14.0-37-generi Thu Apr 16 05:51   still running
admin    tty2         tty2             Wed Apr 15 19:28 - crash  (10:23)
admin    seat0        login screen     Wed Apr 15 19:28 - crash  (10:23)
reboot   system boot  6.14.0-37-generi Wed Apr 15 19:25   still running
admin    tty2         tty2             Mon Apr 13 03:10 - crash (2+16:15)
admin    seat0        login screen     Mon Apr 13 03:10 - crash (2+16:15)
```

Conclusion:

The lab experiment provided a clear understanding of Linux operating system commands and their practical applications. Various commands related to file and directory management, permission control, system monitoring, and compression were successfully practiced in the Ubuntu environment. This hands-on work improved understanding of both basic and advanced Linux commands and their real-life use. Overall, the lab enhanced practical skills, problem-solving ability, and familiarity with Linux terminal operations, which will be useful for future academic and professional tasks.

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